ONPG Bead Reactor, Enzyme Kinetics and Internal Diffusion

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Simulation: https://farazrahimi95-collab.github.io/labs/onpg/

Development date: _____

Name(s): _____ Date: _____

Educational Purpose

Through this virtual experiment students will examine β-galactosidase enzymatic ONPG hydrolysis, compare results for two immobilized-enzyme bead sizes, and use free-enzyme reaction rate data to study Michaelis-Menten kinetics.

Student Learning Objectives

• Use calibrated absorbance readings at 415 nm to determine ONP concentration. Note ONP has an absorbance maximum for light at 415 nm.

• Compare ONP production for 2.8 mm and 4.6 mm beads prepared with the same reactor total enzyme charge.

• Determine an initial reaction rate from the slope of an ONP-concentration-versus-time line.

• Use six initial rates to determine the maximum reaction rate V_{max} and the Michaelis-Menten constant, K_M .

• Understand how bead diameter affects the internal effectiveness of the enzymatic ONPG reaction.

System and Measurement

Reaction and optical measurement

β-galactosidase cleaves the substrate OrthoNitroPheno-Galactose (ONP-G) bond to transition a non-colored ONPG to yield free galactose and the yellow colored ONP.

A spectrophotometer measures ABS_{415} related to how much light the ONP absorbs at 415 nm.

Blanking sets the spectrophotometer baseline to zero so background absorbance from the cuvette and solution is not attributed to ONP absorbance in the solution.

$$\mathrm{ABS}_{415} = 3.58 \times C_{ONP}$$

$$C_{ONP} = \mathrm{ABS}_{415} / 3.58$$

ONP concentration is reported in [mM]; absorbance is dimensionless.

Before Operating the Simulation

The recirculating apparatus is shown in Fig. 1. The two laboratory bead sizes are shown in Fig. 2.

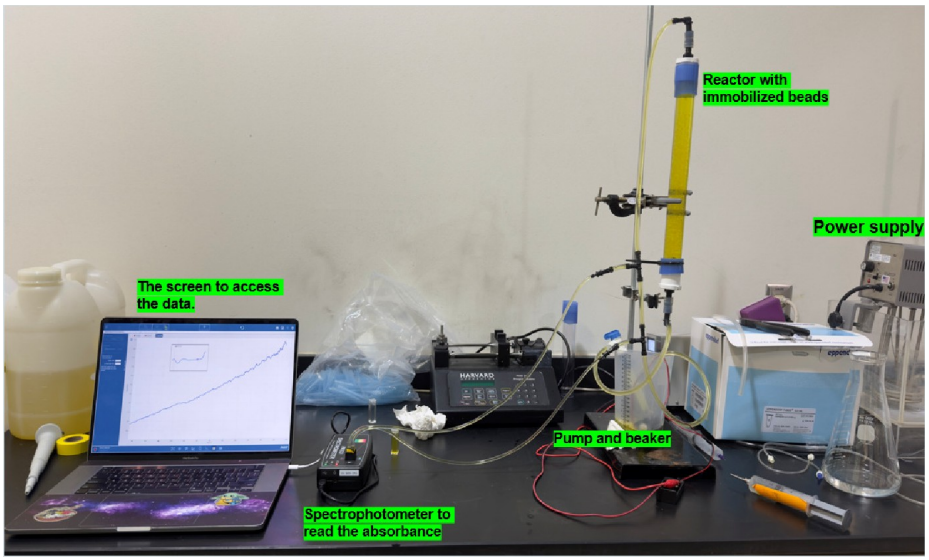


Fig. 1 Recirculating immobilized-enzyme apparatus used to motivate the Stage 1–3 simulation.

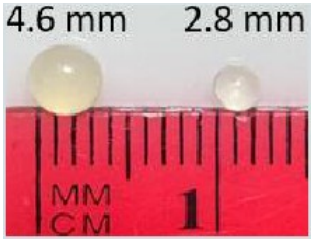


Fig. 2 Approximate bead sizes.

Click the About button at the top of the ONPG simulation and read the short system description. The pump recirculates liquid from the reservoir through the fluidized bed of beads containing immobilized β-galactosidase enzyme, routing fluid through the in-line spectrophotometer, and back to the reservoir. Stages 1 and 2 use the same total enzyme charge and initial ONPG concentration; only the bead diameter changes.

System and Measurement Continued

Before beginning a run, the experimentalist obtains a baseline absorbance for the clear liquid containing ONPG and zeroes the spectrophotometer. This subtracts background absorbance from all samples so that ABS_{415} measures ONP alone.

The cuvette series in Fig. 3 illustrates increasing ONP color. Use the calibration relationship in Fig. 4 to convert absorbance to concentration.

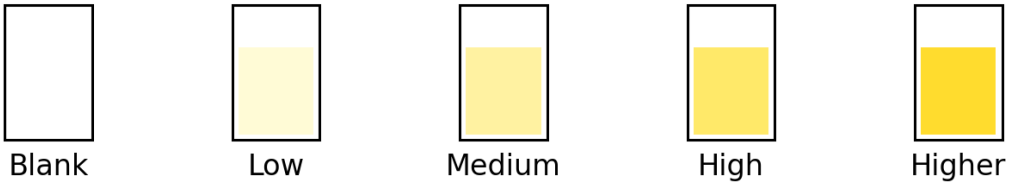


Fig. 3 Qualitative cuvette series with increasing ONP content. Colors are illustrative, not measured standards.

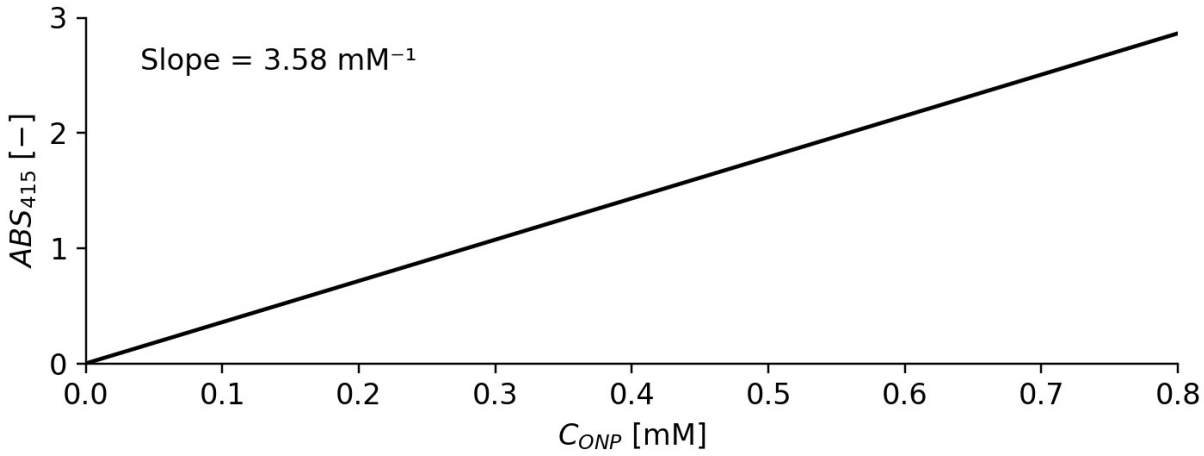


Fig. 4 Supplied calibration relationship, ABS_{415} = 3.58 × C_{ONP}. The line is calculated from the supplied equation; measured calibration points were not provided.

$C_{ONP} = \mathrm{ABS}_{415} / 3.58$

Predictions Before Experiment 1

Before running the simulation, which bead size do you expect to produce ONP faster, the 2.8 mm or 4.6 mm bead? Give reasons for your prediction.

Before running the simulation, sketch C_{ONP} versus time for both bead sizes on Fig. 5.

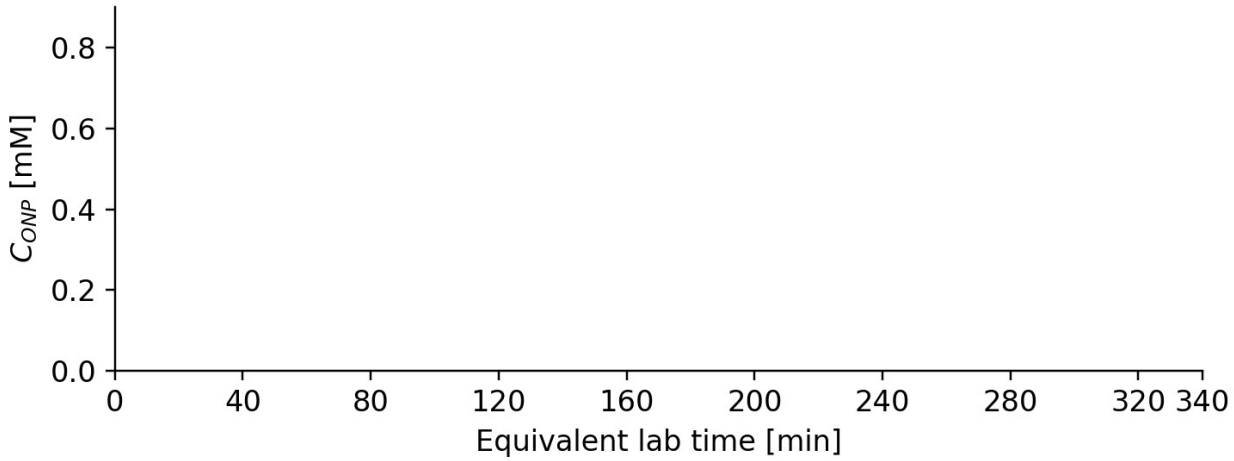


Fig. 5 Prediction axes for 2.8 mm and 4.6 mm beads. Label both curves.

What are the reasons you sketch this the way you did?

Experiment 1 Compare the Two Bead Sizes Stages 1–3

Purpose: Compare C_{ONP} vs. time for 2.8 mm and 4.6 mm beads.

Procedure

1. Open Stage 1 (2.8 mm beads). Click “Set autosampler · every 28.33 min.” The interval is expressed in equivalent lab minutes.

2. Click “Start 340-min run (2-minute playback).” Record all 13 readings. The 120 s playback represents 340 equivalent lab min.

3. Continue to Stage 2 (4.6 mm beads), set the same automatic sequence, and run the simulation.

4. Open Stage 3 to compare the two saved runs. No new reactor run is performed in Stage 3.

Experiment 1 Paired Results

Record the paired results from Stage 3 below. Both the simulation and this table use equivalent lab time in minutes.

A 2-minute playback represents 340 equivalent lab minutes. The autosampler records 13 readings, including the zero baseline.

Equivalent lab time [min]	C_{ONP} [mM] 2.8 mm bead	C_{ONP} [mM] 4.6 mm bead
0	0	0
28.33		
56.67		
85		
113.33		
141.67		
170		
198.33		
226.67		
255		
283.33		
311.67		
340		

At the same time points, which bead size produced the higher ONP concentration? Cite one pair of values from the table, e.g., at 170 min.

Experiment 1 Bead-Size Comparison and Internal Mass Transfer

Use the Stage 3 comparison screen and record the summary values.

Bead diameter [mm]	Initial rate [mM/min]	Time to 50% [min]	Time to 80% [min]	Final conversion [%]	η_i [–]
2.8					
4.6					

Which bead preparation reacted faster? Support your answer with the initial rate and one time-to-conversion value.

Experiment 1 Internal Concentration and Effectiveness

The internal effectiveness factor, η_i, is the actual whole-bead reaction rate divided by the rate that would occur if the entire bead were at the bead-surface substrate concentration. A lower η_i indicates a stronger internal-diffusion limitation.

Smaller beads have more external area per total bead volume, but internal substrate availability determines the diffusion limitation. ONPG is consumed in the outer region of a bead; if diffusion cannot replenish it quickly enough, the inner region receives little ONPG and contributes less to the overall reaction rate.

Which bead has the larger decrease in ONPG concentration with dimensionless radius, i.e., from the bead surface to the bead center?

_____

_____

What is happening to ONPG inside the larger bead compared to the smaller bead? Look at $r / R = 0.6$. Hints:

What is the concentration of the ONPG for the larger bead at $r / R = 0.6$ compared to the smaller bead?

If the concentration in one case is higher what happens to the reaction rate inside the bead at the same r / R? Consider mass action effects where larger concentrations mean more frequent molecular collisions.

If ONPG concentration is very low below $r / R = 0.6$ what happens to the reaction rate there?

How is that difference reflected in η_i?

_____

_____

_____

Return to your prediction before Experiment 1. What would you keep or revise after comparing the two runs?

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_____

Experiment 2 Free-Enzyme Kinetics Stage 4

Purpose: determine the initial rate at six initial ONPG concentrations while holding enzyme amount, solution volume, pH, and temperature constant.

Why the slope is used

Each trace plots ONP concentration vs. time. During the first few seconds, the response is treated as approximately linear. The slope of each line is the initial reaction rate.

$v_0 = \Delta C_{ONP} / \Delta t$

The six slopes become six points on the Michaelis–Menten graph in Stage 5, where v_0 is plotted against initial ONPG concentration.

Procedure

1. Open Stage 4 and click “Load six cuvettes.”

2. Click “Blank spectrophotometer.” The background reading is subtracted and the baseline becomes zero before further color change due to yellow-colored ONP formation.

3. Click “Start 4.5-s scan.” All six ONP-concentration-versus-time lines are generated on the same graph.

4. Click “Show initial rates” and record the six slopes below.

Initial ONPG concentration, $C_{ONPG,0}$ [mM]	**Initial rate, v_0 [mM/s]**
0.622	
1.24	
2.49	
4.98	
9.95	
19.9	

Compare the slope at 0.622 mM ONPG with the slope at 19.9 mM ONPG. What does this tell you about how initial rate changes as substrate concentration increases? Considering how molecules interact as a function of concentration explain why the rate increases significantly.

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Experiment 2 Michaelis–Menten Analysis Stage 5

1. Before opening the Stage 5 fitted curve, predict the shape of v_0 versus $C_{ONPG,0}$. Sketch it on Fig. 6 and explain why you expect this behavior.

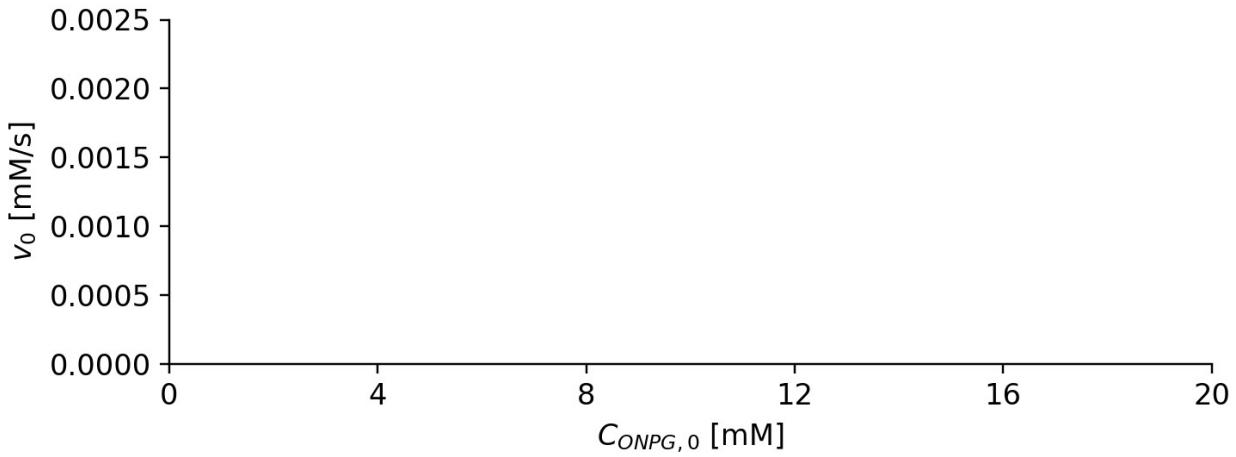


Fig. 6 Prediction and measured initial-rate plot. Use a dashed prediction curve and label the six Stage 4 points.

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2. Plot your six actual Stage 4 initial rates on Fig. 6. Then build the Stage 5 kinetics plot and compare it with your prediction.

$$v_0 = \frac{V_{max}\ C_{ONPG,0}}{K_M + C_{ONPG,0}}$$

3. Linearize the Michaelis–Menten equation by plotting $1/v_0$ against $1/C_{ONPG,0}$. Use your six measured rates and a straight-line fit to determine the slope and intercept. Prepare the reciprocal plot on a separate sheet or in a spreadsheet.

$$1/v_0 = (K_M / V_{max})(1/C_{ONPG,0}) + 1/V_{max}$$

$$V_{max} = 1 / intercept$$

$$K_M = slope / intercept$$

Slope [s]: _____ Intercept [s/mM]: _____

V_{max} [mM/s]: _____ K_M [mM]: _____

4. Check your reciprocal fit: V_{max} ≈ 0.00188 mM/s and K_M ≈ 1.17 mM (slope ≈ 624 s; intercept ≈ 533 s/mM). The supplied Stage 5 curve instead uses V_{max} = 0.00226 mM/s and K_M = 2.00 mM. These are different fitting results; the difference is not just rounding.

Experiment 2 Interpretation of the Kinetics

Describe the overall shape of v_0 versus initial ONPG concentration.

At low C_{ONPG} values the responses increase linearly with concentration. What is happening between the ONPG molecules and the enzyme molecules that make this occur?

When $C_{ONPG,0}$ >> K_M, to what value does v_0 approach? You will see it is V_{max}. Now explain what is happening. Think about how the enzyme works – ONPG binds to the enzyme, is held momentarily and then the enzyme converts it to colored ONP which takes time. So, what's happening at very high C_{ONPG} values to make the reaction rate to be constant?

Calculate V_{max} / 2. On the Stage 5 graph, locate this rate and estimate the corresponding initial C_{ONPG}. How does that concentration compare with K_M?

Meaning of the Michaelis–Menten Parameters

For the initial-rate mechanism below, E is free enzyme, S is ONPG, ES is the enzyme–substrate complex, and P is product. Assume negligible reverse reaction from product and an approximately steady ES concentration.

$$E + S \rightleftharpoons ES \rightarrow E + P$$

The association, dissociation and product-formation rate constants are k_1, k_{-1} and k_2, respectively.

$$0 \approx d[ES]/dt = k_1[E][S] - (k_{-1} + k_2)[ES]$$

$$[E]_T = [E] + [ES]$$

$$K_M = (k_{-1} + k_2) / k_1$$

$$[ES] = [E]_T[S] / (K_M + [S])$$

$$v_0 = k_2[ES]; \quad V_{max} = k_2[E]_T$$

Thus K_M is the substrate concentration at half the maximum initial rate. It is not generally an equilibrium constant: only when product formation is slow relative to complex dissociation ($k_2 \ll k_{-1}$) does K_M approximate the dissociation constant k_{-1}/k_1. V_{max} is the limiting rate at substrate saturation for the given total enzyme concentration.

What does K_M represent for this enzyme–substrate system, and what does V_{max} represent?

Reference: H. S. Fogler, Elements of Chemical Reaction Engineering, 5th ed., Chapter 9, Enzymes. https://websites.umich.edu/~elements/5e/09chap/summary.html

Experiment 3 Bead-Diameter Design Stage 6

Purpose: vary bead diameter and use the internal effectiveness factor to identify the largest tested bead diameter that meets $\eta_i \geq 0.95$.

Procedure

1. Open Stage 6. Change bead diameter using the slider or the − / + controls.

2. Near the $\eta_i = 0.95$ boundary, change diameter in 0.1 mm increments and click “Save this result.”

3. Save one result immediately below $\eta_i = 0.95$ and the largest tested diameter at or above $\eta_i = 0.95$.

4. Compare the two saved surface-to-center ONPG concentration profiles.

Result	Bead diameter [mm]	η_i [−]	Surface ONPG [mM]	Center ONPG [mM]
Large laboratory bead	4.6	0.290		
Small laboratory bead	2.8	0.450		
Closest saved result below 0.950				
Largest saved diameter at or above 0.950				

At $\eta_i \geq 0.95$ what is true about the concentration profile?

What does that say about diffusion vs. reaction rate for these conditions? Think about the driving force for diffusion, a concentration gradient, and the driving force for reaction, a specific concentration in the M-M equation.

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Which bead diameter would you select from your tested values? Use the two ONPG concentration profiles and the internal effectiveness factors to support your answer.

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Would there be any detriments to choosing a very small bead diameter?

_____

Why should we immobilize the enzyme at all if the fastest rate is when we have no beads at all, just the enzyme in solution.

_____

Report numerical answers with units where applicable. Use values from the simulation tables and graphs when the question asks for evidence.